

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: VIII Semester B.E. Computer Science and Engineering.

Course Code & Title: CS6801 Multi-core Architecture and Programming

Name of the Faculty member: M.Gomathy Nayagam, Asst.Professor (SG)/CSE

Innovative practice: Cross Word Puzzle

Type of Learning: Active Learning

Topic: Symmetric and Distributed Memory Architecture and Cache Coherence

Date: 07.01.2020

Learning Objectives:

Objective(s)	Statements
O1	To expose the students to describe the symmetric and distributed memory architecture, cache coherence concepts.
O2	To make the students to function effectively as an individual and member in a team to solve the brainstorming session

Justification for the Topic:

To identify the understanding level of a student in symmetric & distributed memory architecture and cache coherence topic, a game based activity was chosen to do the revision of topic instead of carrying out regular quiz tests.

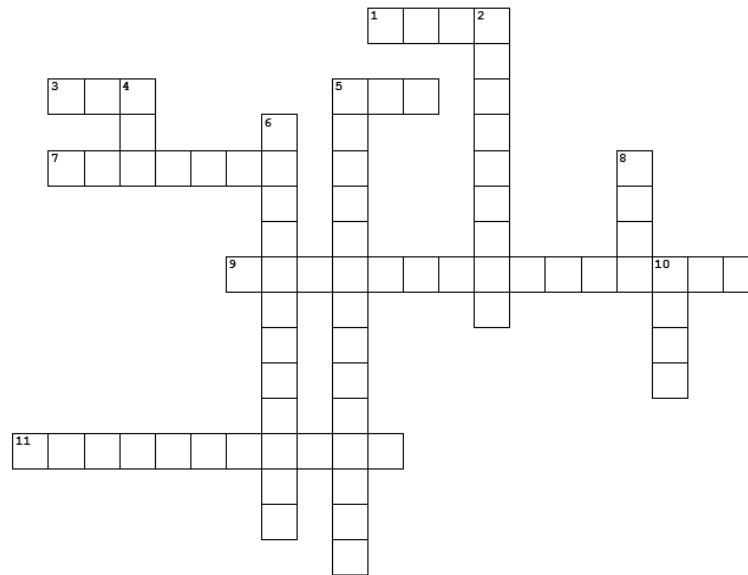
Use of Cross word Puzzle:

- It helps the students to engage deeply to understand the topic.
- It motivates the students to overcome the boredom to study the concepts.
- It helps the students to keep their brain active and sharp to recollect the all the term related to the topic.

Activity:

- The entire class is divided as a group with two members.
- Crossword puzzles of 10 questions were given to each group of students. The crossword question puzzles given to the students are shown in fig.1.
- Time allotted for the activity is 15 minutes.
- The group members were discussed among themselves and answered the question in the paper within 15 minutes and returned the answer script to course instructor.
- Course instructor corrected the puzzle answers and the answers of wrong questions were discussed at the end of the class to make them to know the answer of the entire questions. The sample answers were shown in figure.2.

Symmetric and Distributed Shared Memory Architectures, Cache coherence



Across

1. Multiple simultaneous instruction streams operating on multiple data stream(4 Letters)
3. Time to access all the memory location will be same for all the core architecture (3 letters)
5. Communication wires are shared by the devices called (3 letters)
7. A collection of commodity systems connected by interconnection networks (7 Letters)
9. A processor that is able to process sequences of data with a single instruction (16 letters with 2 words)
11. the boss of CPU(11 letters with 2 words)

Down

2. data structure used in directory based cache coherence is (9 Letters)
4. the worker of CPU(3 Letters)
5. a number of simultaneous connection or connectivity in toroidal mesh interconnects. (14 letters with 2 words)
6. data is written in the cache and the corresponding main memory location at same time is called (12 letters, 2 words)
8. Language provide some of the mechanism of shared memory programs.
10. Applies the same instruction to multiple data items or it's idle (4 Letters)

Fig 1: Crossword Puzzle given to the students

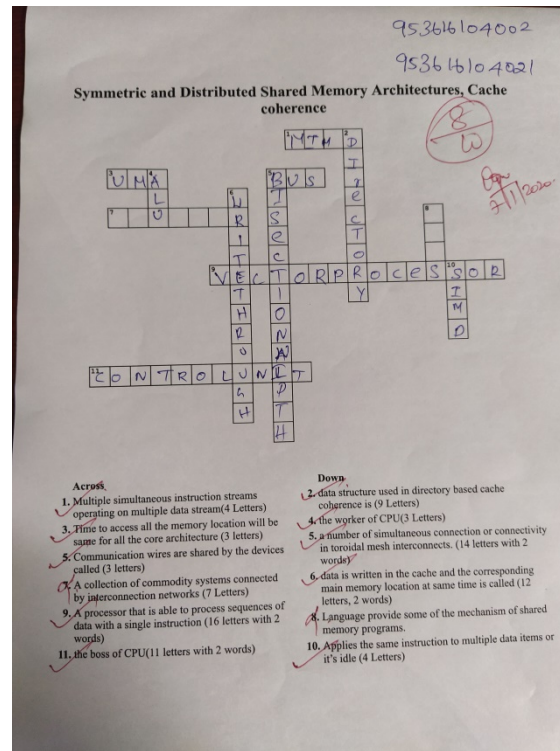


Fig 2: Sample Crossword Puzzle answer of Group 1(Ms.Ashaya.M & Ms.Jesima.A.S.)

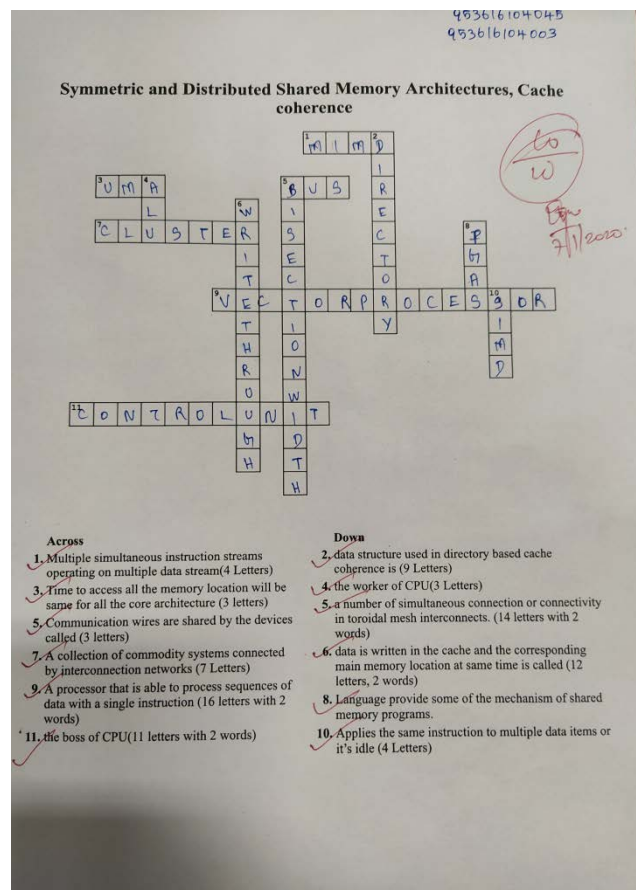


Fig 2: Sample Crossword Puzzle answer of Group 2(Ms.P. Amirtha & Ms.S.Selvakumari)

Outcome:

Based on the answers provided by the students, analysis was performed and it is given as a chart in figure 3. From that we can find out that, most of the students had familiarity with the basic concepts of symmetric, shared memory architecture and cache coherence. This activity helps the students to recall important key terms easily, think individually and improve their problem solving skills through team work.

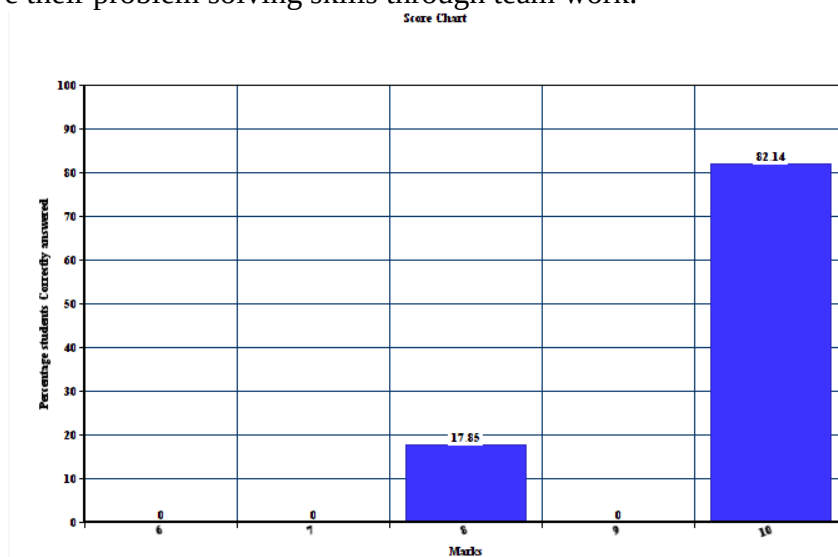


Fig.3 Score Analysis for Crossword Puzzle

Objective and PO Mapping

Objective(s)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	1	1	-	-	-	-	2	-	1	-	-
O2	1	1	-	-	-	-	-	-	3	2	-	2

Reflective Critique:

Identified Challenges:

- Student felt difficult by solving the crossword puzzles on system.
- Few students wrote wrong keywords on the puzzle for particular question.

Addressing Challenges:

- Faculty given hardcopy of cross word puzzle sheet to individual groups. So, they could write their answer in the hard copy.

Post Implementation:

- They easily remembered the concepts and key terms related to symmetric, shared memory architecture and cache coherence topic.
- Faulty discussed answer for all questions to know the correct answer for all the questions by the students.

References:

1. Ms.P.Jothi Thilaga, "Files and Exception" in course "GE8151 Problem Solving and Python Programming" dated:19.10.2019. (https://www.ritrjpm.ac.in/images/computer-science/48.GE8151_Crossword.pdf)
2. Mrs.B.Vijayalakshmi "Linear Data Structures - List" in course "CS8391 Data Structures" dated: 15.07.2019. ([https://www.ritrjpm.ac.in/images/computer-science/DS-Crossword% 20Puzzle.pdf](https://www.ritrjpm.ac.in/images/computer-science/DS-Crossword%20Puzzle.pdf))

3. <https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners>
4. <https://voxy.com/blog/2011/05/are-flashcards-an-effective-learning-tool-infographic/>
5. <https://elearningindustry.com/the-5-best-free-flashcard-creation-tools-for-teachers>
6. <https://www.education.com/worksheet-generator/reading/crossword-puzzle/>

Faculty in-charge

HOD/CSE